

Anna Dorosh

MSc Student | Materials Science | AI in Chemistry & Catalysis

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PROFILE

MSc student in Materials Science at the Skolkovo Institute of Science and Technology with a background in biotechnology, pharmaceutical engineering, and data analysis. Research interests focus on artificial intelligence and machine learning in chemistry and materials science, particularly catalysis, prediction of chemical properties, and data-driven materials discovery. Experimental experience includes catalytic biomass conversion, laboratory-scale process studies, and quantitative data analysis.

EDUCATION

2026–Present	Master of Science in Materials Science Skolkovo Institute of Science and Technology (Skoltech), Moscow, Russia Interdisciplinary training in materials science, chemistry, and computational approaches.
2022–2026	Bachelor of Science in Biotechnology ITMO University, Saint Petersburg, Russia GPA: 4.75/5.00. Academic background in analytical and physical chemistry, bioinformatics, pharmaceutical technologies, and experimental research.
Additional	Professional Retraining Diploma — Data Analysis Technologies ITMO University, Saint Petersburg, Russia Training in programming, data processing, statistical analysis, visualization, and computational methods.
Feb–Jun 2025	Exchange Semester — Pharmaceutical Engineering Nanjing University of Science and Technology (NJUST), Nanjing, China English-taught academic exchange semester in Pharmaceutical Engineering.

RESEARCH EXPERIENCE

Sep–Dec 2024

Catalytic Pyrolysis of Wood Biomass

Laboratory Research Project, ITMO University

- Conducted laboratory experiments on the catalytic pyrolysis of wood biomass.
- Prepared biomass samples and metal-containing catalytic systems for pyrolysis experiments.
- Produced bio-oil and investigated how metal salt additives affected the yield and distribution of pyrolysis products.
- Analyzed experimental results related to thermochemical biomass conversion and catalytic performance.

Sep 2025–Mar 2026

Research Intern

ANO “Biosphere Technologies” / All-Russian Institute of Plant Protection, Russia

- Investigated the bioconversion of grain-production waste using *Hermetia illucens* larvae.
- Prepared waste-based substrates and conducted laboratory-scale cultivation experiments under controlled conditions.
- Evaluated the efficiency of conversion of grain waste generated during *Sitotroga cerealella* egg production.
- Collected and analyzed experimental data to assess process performance and approaches to waste valorization.

RESEARCH INTERESTS

- Artificial intelligence and machine learning in chemistry
 - Computational chemistry
 - Data-driven materials discovery
- Data-driven catalyst discovery and optimization
 - Prediction of chemical and physicochemical properties

SKILLS

Programming & Data

- Python
- Data preprocessing and statistical analysis
- Scientific data visualization
- Fundamentals of machine learning

Scientific Work

- Experimental design
- Scientific literature review
- Data analysis and interpretation

Chemistry & Experimental

- Analytical and physical chemistry
- Catalytic pyrolysis
- Sample preparation
- Spectrophotometry and basic HPLC

Languages

- Russian — Native
- English — C1 / Advanced